Ryland Gross

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Education

Imperial College London London, United Kingdom
MSc in Pure Mathematics *Expected June 2027*

Harvard University Cambridge, MA
A.B. *cum laude* in Mathematics May 2026
Citation in Russian Language

Research Interests

Commutative algebra; combinatorics; formal verification of mathematics; automated theorem proving; AI for math.

Research Experience

Autoformalization of Commutative Algebra in Lean 4 2025–present
Independent Research (GitHub)

Developing a Lean 4 / Mathlib formalization of results from Atiyah and Macdonald's *Introduction to Commutative Algebra*. Current work includes formalization of results in ring theory, ideals, modules, and dimension theory. The formalization is also being used to construct a benchmark for evaluating AI-assisted theorem proving in commutative algebra.

Harvard Mathematics REU May–August 2025
Research Investigator, supervised by Dr. Philip Matchett Wood Cambridge, MA

Conducted research in combinatorial matrix theory, focusing on polytope varieties, Birkhoff polytope classes, and near-derangement matrices. Developed a depth-first search algorithm for maximizing trace over this matrix class and obtained new lower bounds for exclusive near-derangement total support matrices trace.

ArcGIS Spatial Analysis, Kentucky Geologic Survey May–August 2023
Research Intern Lexington and Hazard, KY

Collaborated with the Kentucky Geologic Survey and the Housing Development Alliance on GIS-based methods for identifying viable housing sites for Eastern Kentucky flood-displacement victims. Identified 15 candidate building sites; 2 homes were subsequently constructed.

Software and Presentations

- **LeanTeX** (GitHub) — A LaTeX package and Python command-line tool for embedding Lean 4 proof snippets and Infoview output into mathematical documents.
- *GIS-Based Housing Site Identification for Eastern Kentucky Flood Victims*. Appalachian Research Conference, 2023.

Teaching Experience

Harvard University Cambridge, MA
Course Assistant September 2023–May 2026

Served as course assistant for three undergraduate courses:

- Math 22b: Linear Algebra and Vector Analysis
- Math 130: Classical Geometry (Euclidean and Non-Euclidean)
- Math 157: Mathematics in the World (Head CA)

Taught students topics such as linear algebra, multivariable calculus, Euclidean and non-Euclidean geometry, introductory probability, elementary complex analysis, cryptography, Python programming, and proofs.

Outreach and Service

Prison Mathematics Project — Remote / Cambridge, MA
Mentor — October 2025–April 2026

Provided one-on-one mathematical mentorship to an incarcerated individual, covering topics from elementary algebra and calculus through number theory, proof-writing, and abstract algebra.

Community Web Development, Eastern Kentucky — 2025–present
Volunteer

Designed and maintained websites for local churches, nonprofits, and small businesses in Eastern Kentucky. Recent projects include a memorial foundation website for the Team Colton DIPG Foundation supporting pediatric brain cancer awareness.

Graduate and Advanced Coursework

Math 221: Graduate Commutative Algebra

Math 243: Graduate Evolutionary Dynamics

Math 161: Formal Methods in Mathematics (Lean 4 / Mathlib)

Math 114: Real Analysis II

Math 124: Number Theory

Math 131: Topology and the Fundamental Group

Math 113: Complex Analysis

Math 136: Differential Geometry

Math 122: Abstract Algebra I and II

Statistics 111: Introduction to Statistical Inference

Statistics 110: Probability Theory

Technical Skills

Proof Assistants: Lean 4 / Mathlib
Programming Languages: Lean 4, Python, Rust, R
Tools: Git, LaTeX, ArcGIS

Citizenship and Languages

Citizenship: United States
Languages: English (native), Russian (reading)